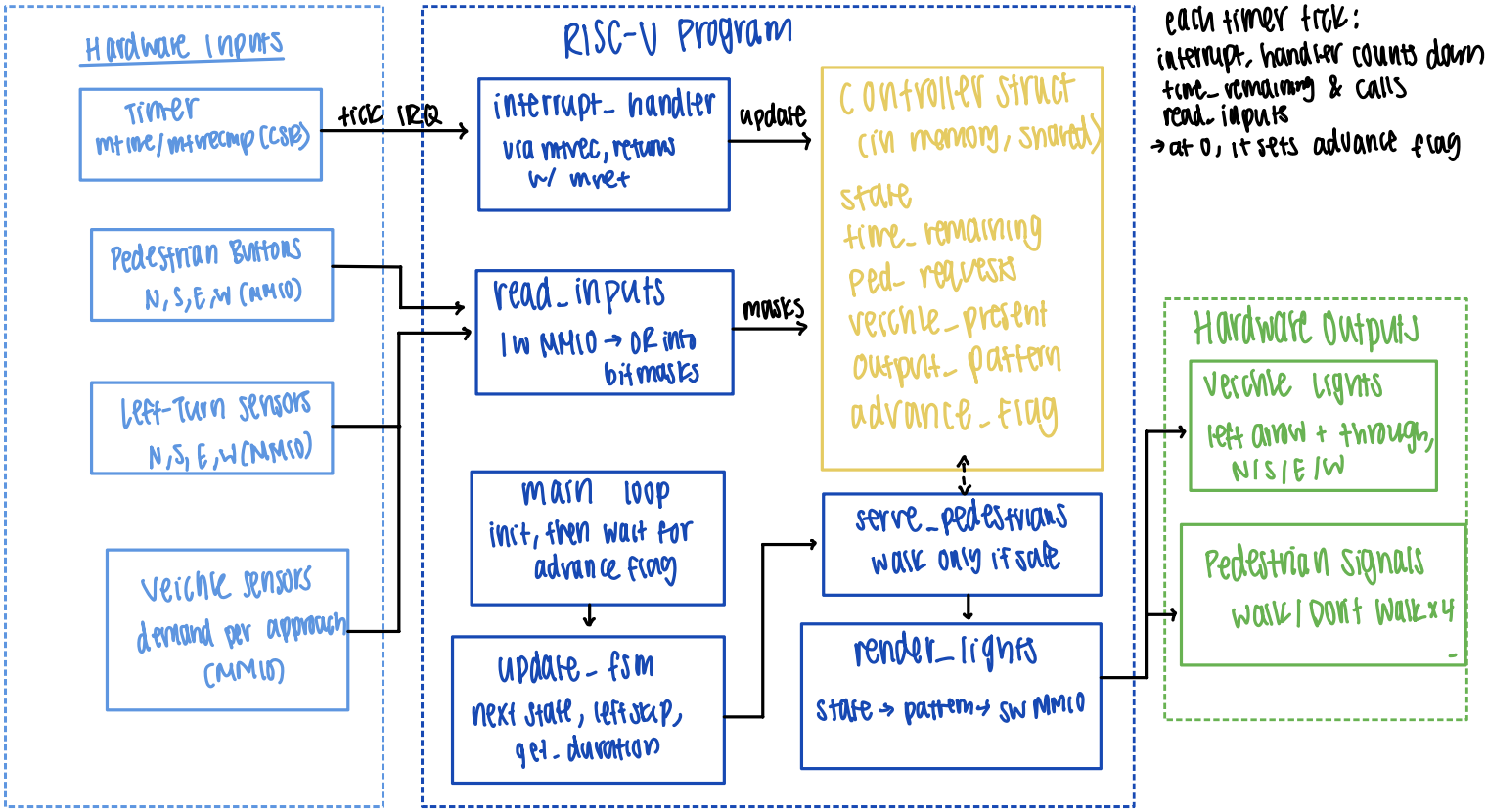

Hardware Inputs
Timer
mtime/mtimecmp (CSR)
tick IRQ
Pedestrian Buttons
N, S, E, W (MMIO)
Left-Turn Sensors
N, S, E, W (MMIO)
Vehicle Sensors
demand per approach
(MMIO)
RISC-V Program
interrupt_handler
via mtvec, returns
w/ mret
update
read_inputs
lw MMIO → OR into
bitmasks
masks
main loop
init, then wait for
advance flag
update_fsm
next state, left_stop,
get_duration
Controller struct
(in memory, shared)
state
time_remaining
ped_requests
vehicle_present
output_pattern
advance_flag
serve_pedestrians
walk only if safe
render_lights
state → pattern → sw MMIO
each timer tick:
interrupt handler counts down
time_remaining & calls
read_inputs
→ at 0, it sets advance flag
Hardware Outputs
Vehicle Lights
left arrow + through,
N/S/E/W
Pedestrian Signals
walk / don't walk x4